

HUNTSVILLE EXECUTIVE, AL, USA

WMO: 720307

Lat: 34.861N

Lon: 86.557W

Elev: 755

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63804

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	22.1	3.4	6.7	28.6	9.3	9.1	30.5	22.7	48.9	19.9	42.1	5.1	0	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	95.0	73.9	91.4	73.0	90.3	72.5	77.6	87.8	76.6	86.7	75.5	85.3	5.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.9	134.6	81.8	73.4	127.6	80.1	72.7	124.7	79.5	41.5	88.7	40.4	86.5	39.4	85.7	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	16.6	14.4	12.5	97.5	5.0	3.6	8.9	100.1	6.0	102.1	3.1	104.1	-0.5	106.7
			10.9	78.6	4.6	2.2	7.6	80.2	4.9	81.4	2.3	82.7	-1.1	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	40.7	44.5	51.6	60.4	68.4	76.7	78.7	77.8	72.4	61.0	49.8	44.1
	DBStd	15.70	10.77	10.26	9.49	8.11	7.06	4.03	3.88	4.70	5.92	8.40	8.55	9.54
	HDD50	1006	328	210	98	14	1	0	0	0	0	11	106	238
	HDD65	3286	755	576	423	180	50	0	0	1	13	179	461	648
	CDD50	4869	38	56	147	326	572	800	891	862	671	352	99	55
	CDD65	1678	0	3	7	43	157	350	426	398	233	55	5	1
	CDH74	15745	0	11	64	424	1488	3366	4034	3689	2092	552	24	1
	CDH80	6429	0	3	3	67	467	1466	1806	1672	807	136	2	0
Wind	WSAvg	4.9	6.3	6.7	6.7	6.1	4.5	3.7	3.3	2.8	3.2	4.3	5.1	5.9
Precipitation	PrecAvg	55.6	5.8	5.4	5.3	4.8	4.2	3.9	5.2	3.6	3.7	3.0	4.2	6.5
	PrecMax	70.4	11.3	10.6	9.5	9.9	10.2	7.6	10.7	8.0	7.8	7.2	7.4	17.8
	PrecMin	31.6	2.8	2.7	1.4	2.0	0.6	1.0	2.2	1.0	0.4	0.0	1.7	2.4
	PrecStd	8.7	2.0	2.3	2.0	2.0	2.5	2.1	1.8	1.4	2.2	2.0	1.6	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.4	74.6	79.5	85.6	90.9	95.5	98.7	99.2	93.8	88.4	77.5	71.7
		MCWB	58.9	64.2	63.3	65.2	72.6	71.4	74.3	73.0	72.3	70.8	62.8	62.0
	2%	DB	64.3	69.8	75.2	81.5	88.1	92.5	94.6	95.1	90.6	83.2	72.2	67.8
		MCWB	57.1	62.2	61.4	63.9	70.4	72.2	74.6	75.0	70.6	67.7	60.4	60.5
	5%	DB	62.6	65.2	72.0	79.2	85.7	90.5	90.9	91.1	88.2	80.6	68.8	63.6
		MCWB	56.2	57.2	58.7	62.2	68.4	71.7	74.0	73.5	70.2	65.1	59.9	57.7
	10%	DB	58.8	61.2	68.2	75.8	81.9	88.3	89.6	89.5	85.6	76.8	64.5	60.6
		MCWB	53.5	53.3	57.4	60.9	66.1	71.3	73.6	72.8	70.1	64.0	56.7	55.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.1	66.0	66.2	69.4	75.0	77.9	79.1	79.0	76.2	73.4	66.9	64.1
		MCDB	65.3	71.8	75.6	80.1	86.6	87.9	88.6	89.6	85.5	82.2	72.0	68.9
	2%	WB	59.2	63.4	63.9	67.4	72.9	76.3	77.9	77.4	74.7	71.7	65.0	61.9
		MCDB	63.2	68.2	71.9	77.5	84.2	86.4	87.8	88.0	84.3	79.8	69.7	65.5
	5%	WB	57.2	59.7	61.8	65.7	71.2	75.0	76.9	76.3	73.3	69.2	62.1	59.2
		MCDB	61.9	63.9	68.5	74.7	81.1	84.8	86.8	86.6	82.5	75.5	66.6	63.1
	10%	WB	54.1	55.6	59.3	63.8	69.3	73.6	75.7	75.0	72.0	66.4	58.9	55.2
		MCDB	57.6	60.1	66.0	71.6	77.2	83.1	85.3	84.7	80.3	73.1	63.5	59.6
Mean Daily Temperature Range	5% DB	MDBR	17.7	19.0	21.2	23.1	21.9	21.2	19.7	20.6	22.4	23.7	22.2	17.6
		MCDBR	20.4	23.5	25.9	25.9	25.2	24.6	22.5	23.9	25.8	27.0	25.4	19.8
		MCWBR	14.9	15.4	13.8	11.1	9.3	7.7	7.4	7.3	8.8	12.0	14.8	14.6
	5% WB	MCDBR	17.7	19.7	20.7	20.7	21.6	20.5	19.5	20.5	21.7	20.5	18.1	17.6
	MCWBR	14.7	15.4	12.7	10.6	9.1	8.1	7.3	7.3	8.8	11.6	13.0	14.9	
Clear-Sky Solar Irradiance	taub	0.320	0.325	0.367	0.398	0.449	0.483	0.521	0.531	0.430	0.379	0.339	0.316	
	taud	2.482	2.466	2.370	2.275	2.179	2.115	2.029	1.990	2.261	2.388	2.446	2.512	
	Ebn at Noon	281	292	286	281	267	257	246	241	263	269	271	276	
	Edh at Noon	27	30	36	42	47	50	54	56	40	33	28	25	
All-Sky Solar Radiation	RadAvg	762	931	1298	1689	1857	1979	1884	1744	1509	1212	882	654	
	RadStd	59	115	123	117	144	152	100	91	133	158	95	61	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254	+176

Nomenclature: See separate page